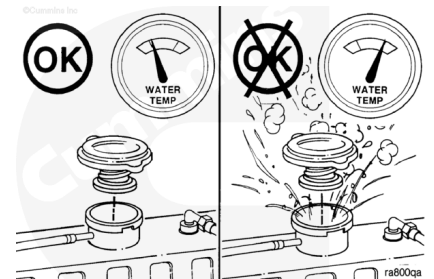


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

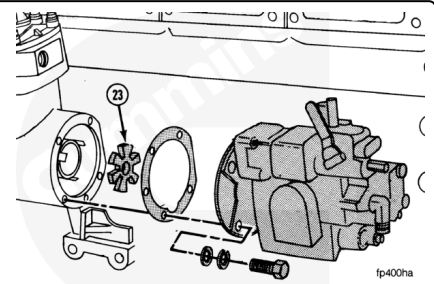


⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the coolant. Refer to Procedure 008-018
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).

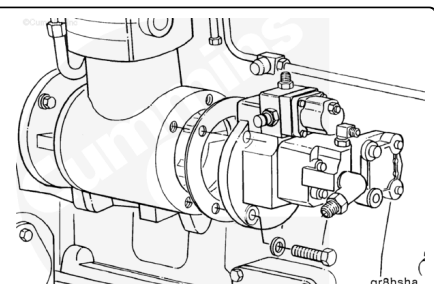
Remove the fuel pump. Refer to Procedure 005-016
(/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).



Remove

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



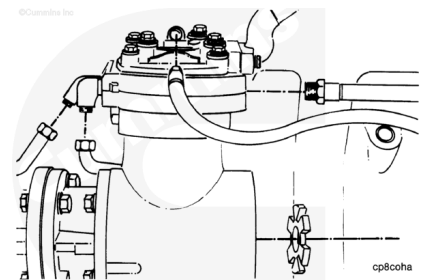
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

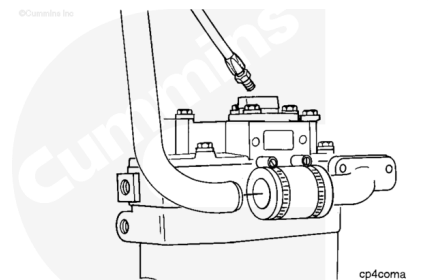
Use steam to clean the air compressor.

Dry with compressed air.

Remove the coolant lines from the air compressor.



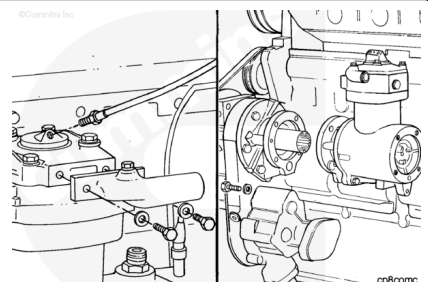
Remove the air inlet and outlet connections from the air compressor.



Single Cylinder

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



Remove the air compressor support bracket and capscrews.

Remove the air compressor mounting flange capscrews.

Remove the air compressor.

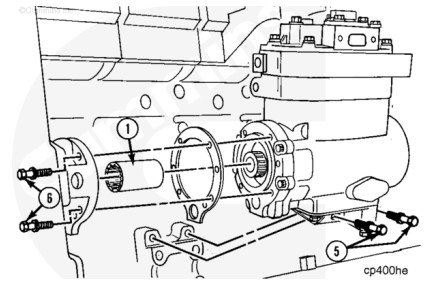
Remove the splined coupling.

Remove and discard the gasket.

Twin Cylinder

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



Remove the air compressor support mounting capscrews (5).

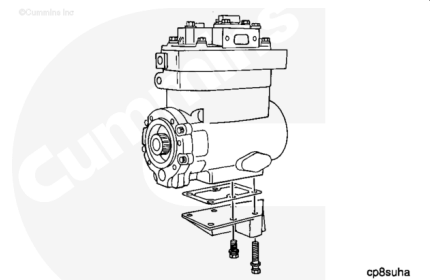
Remove the air compressor mounting flange capscrews (6).

Remove the air compressor.

Remove the splined coupling (1).

Remove and discard the gasket.

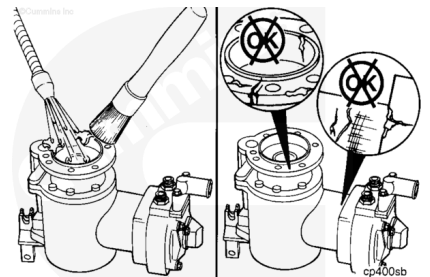
Remove the cover plate/support bracket from the air compressor.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

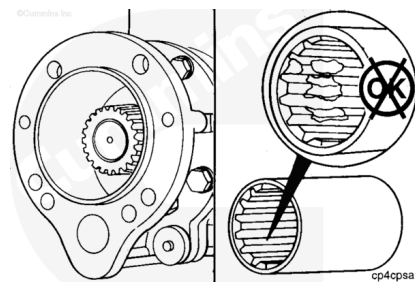
Clean the air compressor with solvent and dry with compressed air.

Inspect the housing for cracks or damage.

If the housing is cracked or damaged it **must** be replaced.

Inspect the splined coupling for cracks or damage.

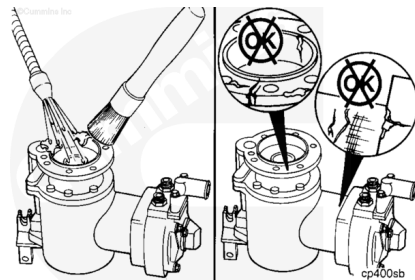
If the splined coupling is cracked or damaged it **must** be replaced.



Single Cylinder

Measure the crankshaft end clearance.

mm		in
0.05	MIN	0.002
0.69	MAX	0.027

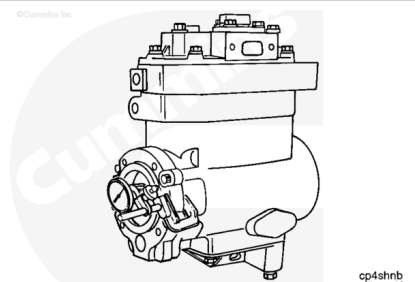


If the crankshaft end clearance is **not** within specifications the air compressor **must** be repaired or replaced. Refer to the Master Repair Manual, Holset® Air Compressors, Bulletin 3666121.

Twin Cylinder

Measure the crankshaft end clearance

mm		in
0.038	MIN	0.001
0.200	MAX	0.008



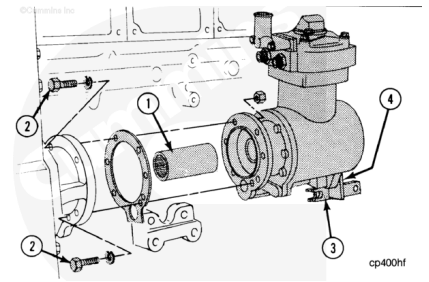
If the crankshaft end clearance is **not** within specifications the air compressor **must** be repaired or replaced. Refer to the Master Repair Manual, Holset® Air Compressors, Bulletin 3666121.

Install

Single Cylinder

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



Install the splined coupling (1) on the accessory drive shaft.

Make sure the accessory drive gasket surface is clean and **not** damaged.

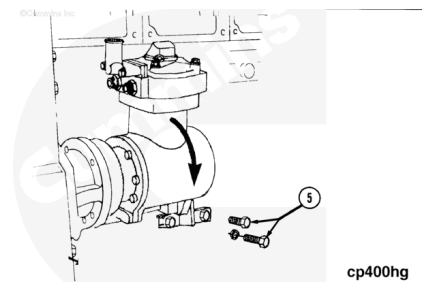
Install a new gasket and the air compressor.

To aid in assembly, do **not** tighten the capscrews until the support bracket is aligned.

Install the four capscrews and nuts (2). The capscrews **must** be loose.

⚠ CAUTION ⚠

The support must be flat against the engine block. If the support is not flat, the compressor and/or accessory drive will fail.



Install the two bracket capscrews (5).

Turn the air compressor until the bracket is flat against the engine block.

Tighten the compressor to accessory drive capscrews (2).

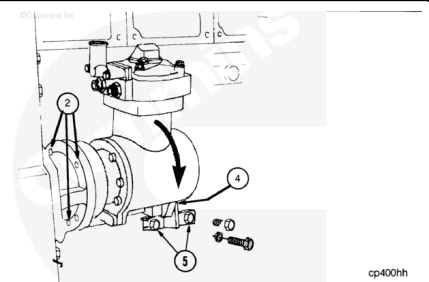
Torque Value: 60 n•m [44 ft-lb]

Tighten the support bracket to engine block capscrews (5).

Torque Value: 45 n•m [33 ft-lb]

Tighten the bracket to air compressor capscrew (4).

Torque Value: 45 n•m [33 ft-lb]



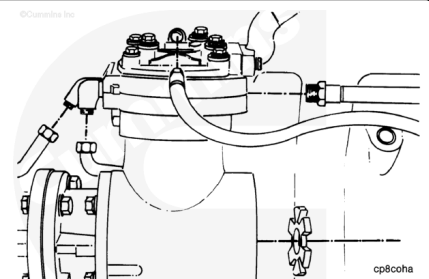
If rubber grommets are used on the coolant lines, make sure they are installed correctly to prevent cuts or tears to the grommets that cause coolant leaks.

When flexible tubing is used, make sure the flexible tube does **not** rub any other surface.

Install the coolant lines to the air compressor and tighten.

Install the air compressor inlet and outline lines.

Tighten the clamps.



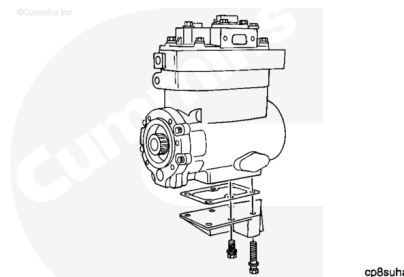
Torque Value: 6 n•m [53 in-lb]

Install the fuel pump drive coupling on the air compressor.

Twin Cylinder

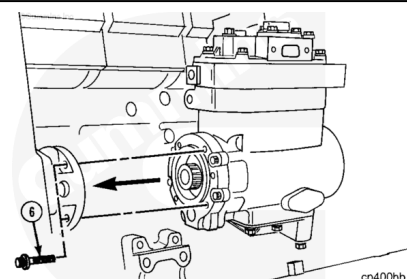
Install the gasket and cover plate/support bracket onto the air compressor.

Do **not** tighten the six capscrews. The support **must** be adjusted.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



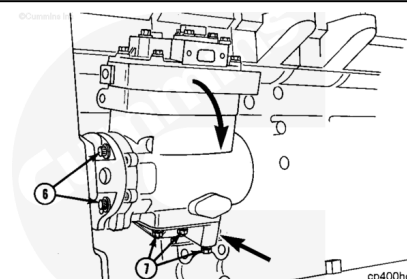
Note : Do not install the splined coupling or the mounting gasket.

Install the compressor on the accessory drive.

Install the four capscrews (6). **Only** tighten the capscrews enough to pull the compressor to the accessory drive. The compressor **must** be turned to align the support bracket.

⚠ CAUTION ⚠

The support must be flat against the engine block. If the support is not flat, the compressor and/or accessory drive will fail.



Tighten the four capscrews (6).

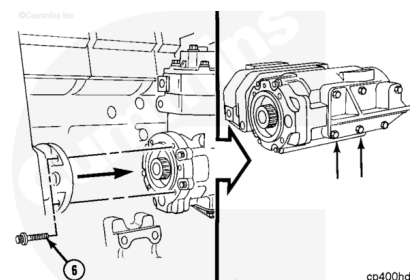
Torque Value: 60 n•m [44 ft-lb]

Tighten the support bracket capscrews (7) that are accessible.

Torque Value: 45 n•m [33 ft-lb]

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



Remove the four capscrews (6).

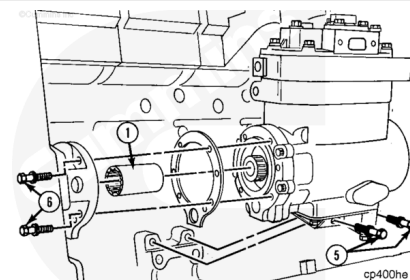
Remove the air compressor.

Tighten the remaining support to air compressor capscrews.

Torque Value: 45 n•m [33 ft-lb]

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury use a hoist or get assistance to lift this component.



⚠ CAUTION ⚠

The support must be flat against the engine block. If the support is not flat, the compressor and/or accessory drive will fail.

Install the splined coupling (1), and the gasket.

Install the air compressor.

Install the four capscrews (6) and two capscrews (5).

Tighten the capscrews.

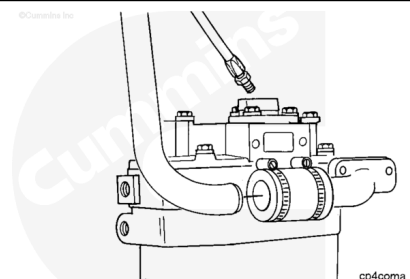
Capscrew (6)	60 n.m	[44 ft-lb]
Capscrew (5)	45 n.m	[33 ft-lb]

Install the fuel pump drive coupling onto the air compressor.

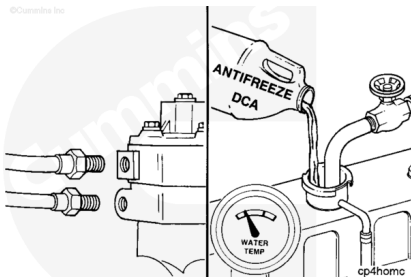
Install the air inlet and outlet connections to the air compressor.

Tighten the clamps.

Torque Value: 6 n•m [53 in-lb]

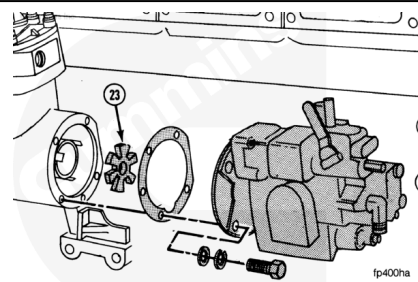


Install the coolant lines into the air compressor.



Finishing Steps

Install the fuel pump. Refer to Procedure 005-016
(</qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html>).



Fill the cooling system. Refer to Procedure 008-018
(</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

Operate the engine to normal operating temperature and inspect for leaks.

